

A comprehensive atlas of AAV tropism in the mouse

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Gene therapy with adeno-associated virus (AAV) vectors requires knowledge of their tropism within the body. Here we analyze the tropism of 10 naturally occurring AAV serotypes (AAV3B, AAV4, AAV5, AAV6, AAV7, AAV8, AAV9, AAVrh8, AAVrh10, and AAVrh74) following systemic delivery into male and female mice. A transgene-expressing ZsGreen and Cre recombinase was used to identify transduction in a cell-dependent manner based on fluorescence. Cre-driven activation of tdTomato fluorescence offered superior sensitivity for transduced cells. All serotypes except AAV3B and AAV4 had high liver tropism. Fluorescence activation revealed transduction of unexpected tissues, including adrenals, testes, and ovaries. Rare transduced cells within tissues were also readily visualized. Biodistribution of AAV genomes correlated with fluorescence, except in immune tissues. AAV4 was found to have a pan-endothelial tropism while also targeting pancreatic beta cells. This public resource enables selection of the best AAV serotypes for basic science and preclinical applications in mice.

INTRODUCTION

Gene therapy requires efficient and specific delivery of the nucleic acid cargo to the tissue and cell type of interest. Over the past three decades, adeno-associated viruses (AAVs) have emerged as the leading viral vector for *in vivo* gene therapy, due to their efficacy and favorable safety profile.¹ AAVs are non-cytopathic and can transduce both dividing and non-dividing cells.² AAV vectors elicit modest and generally manageable immune responses relative to other viral vectors.^{3,4} The recombinant AAV (rAAV) genome forms circular double-stranded episomes, which persist in the nucleus of the target cell. The frequency of chromosomal integrations of rAAV is low, and reported to be in the range of 1%–3% of hepatocytes in the liver, as an example.^{5,6} AAV vectors have been tested in more than 250 clin-

ical trials, culminating in six products approved by the U.S. Food and Drug Administration at the time of this report. Despite this progress, our knowledge of AAV biology remains incomplete. An important goal is to improve the therapeutic utility of AAVs by broadening the range and specificity of tissues and cell types that can be targeted for therapy.

AAVs, members of the *Parvoviridae* family, consist of a 4.7-kb single-stranded DNA genome,⁷ surrounded by a 25-nm diameter non-enveloped protein capsid.⁸ AAVs are dependoviruses that require a helper virus, such as adenovirus or a herpes virus, for replication.⁹ The wild-type AAV genome contains two genes *rep* and *cap*, flanked on either side by approximately 145 nucleotide hairpin like structures called inverted terminal repeats (ITRs).⁷ The *rep* gene encodes four proteins involved in replication, encapsidation, and integration of the viral genome. The *cap* gene produces three different capsid

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proteins, VP1, VP2, and VP3.¹⁰ The VP capsid proteins assemble into an icosahedral structure of 60 total subunits, in a 1:1:10 ratio.¹¹ Three accessory proteins—assembly activating protein,¹² membrane-associated activating protein,¹³ and protein X¹⁴—are also encoded in an alternate open reading frame within *Cap*. rAAVs are generated by replacing the *rep* and *cap* genes with a transgene expression cassette, flanked by the ITRs. In principle any transgene cassette up to approximately 4.9 kb can be packaged into rAAV, enabling treatment of many, but not all, gene therapy targets.

The process of AAV transduction involves multiple sequential events including (1) trafficking to the target tissue, (2) initial interaction with glycans at the cell surface, (3) binding to an internalization receptor, (4) retrograde transport from plasma membrane to the *trans*-Golgi network, (5) escape from the endosome, (6) avoidance of degradation in the cytoplasm, (7) nuclear import, (8) uncoating to release the single-stranded DNA genome, (9) second strand synthesis and formation of circular DNA episomes, and (10) transcription and translation of the transgene.¹⁵ All steps of this complex itinerary are directed by interactions between the capsid and host factors on the cell surface or inside the target cell. Accordingly, the amino acid sequence of the *Cap* gene is the single most important determinant of AAV tropism.

At least 13 distinct AAV serotypes encompassing hundreds of naturally occurring genomic variants have been isolated from multiple vertebrate species.^{16,17} AAV serotypes differ in their *cap* sequence, with the greatest diversity on surface-exposed variable loop regions that participate in receptor binding and antigenicity. AAV's first interaction with target cells involves glycans present on numerous plasma membrane proteins. Glycan binding preferences are known for several serotypes including heparan sulfate proteoglycan for AAV2,¹⁸ N-linked galactose for AAV9,¹⁹ α 2-3 or α 2-6 N-linked sialic acid for AAV5,²⁰ and α 2-3 O-linked sialic acid for AAV4.²⁰ For some serotypes, specific proteins have also been identified as co-receptors, such as α V β 5 integrin for AAV2,²¹ platelet-derived growth factor for AAV5,²² and laminin receptor (LamR) for AAVs2, 3 and 9.²³ Most naturally occurring AAV serotypes, excluding AAV4 and AAVrh32.33,²⁴ require a common receptor called AAV receptor (AAVR) to enter cells.²⁵ AAVR binds to conserved regions in the AAV capsid through its IgG-like polycystic kidney disease domains and facilitates endocytosis.²⁶ Most AAV serotypes also require interactions with GPR108 to escape from the *trans*-Golgi network, with AAV5 as a notable exception.²⁷

The tropism of rAAV vectors *in vivo* is determined by a complex interplay of many factors, including sex, age, dose, route of administration, residence time in the circulation, relative abundance of glycan and internalization receptors, competition between tissues, interaction with the innate and adaptive immune systems, and cell type-dependent pathways. It should be noted that AAV transduction in tissue culture is inefficient for most serotypes and does not predict *in vivo* tropism. Tropism can also vary greatly across species, and even among different strains of the same species. For example, AAV3B is exceptionally poor at transduction of mouse liver, but one of the

best serotypes for human hepatocytes.^{28,29} Likewise, the engineered AAV9 peptide insertion variants AAV-PHP.B which crosses the blood brain barrier following systemic delivery, are effective only in mouse strains expressing the LY6A receptor.³⁰ Despite important species differences, common themes do emerge, which have greatly improved our understanding of AAV biology and vectorology.

The mouse remains the primary model for initial preclinical testing of AAV gene therapies. The vast majority of studies have focused on a limited subset of commonly used human and primate AAV isolates due to their likelihood for effective translation to humans. In a seminal paper, Zincarelli et al.³¹ performed the most extensive analysis of AAV tropism in mice to date. The authors profiled the biodistribution and expression of nine AAV serotypes (AAV1–9) across nine tissues and six serial timepoints following tail vein injection of a luciferase reporter vector. This study revealed a strong tropism for the liver for almost all serotypes, with varying kinetics of expression. The value of such well-controlled comparative studies cannot be understated. However, the study design did not allow for a detailed examination of transduction at the level of individual cells and lacked a broad survey of organs and tissue types. It also focused only on male mice.

In the current work, we performed a comprehensive survey of AAV tropism in male and female mice following systemic administration using fluorescent reporter activation. We define the biodistribution and functional transduction of 10 naturally occurring AAV serotypes across 22 different tissues, including reproductive organs and HSCs. By using fluorescent protein activation as a readout of transduction, we were able to detect rare events and, in some cases, identify specific target cell types. We have identified novel tropisms for multiple serotypes, most notably AAV4, which broadly delivers to endothelial cells in most tissues as well as beta cells in the pancreas. Our results have generated a comprehensive atlas of murine AAV tropism that is freely available to researchers interested in using AAVs as gene delivery vehicles *in vivo*. Data and images associated with these studies are publicly available at <https://scge.mcw.edu/toolkit>. This resource will help in the selection of the best serotype for preclinical gene therapy studies, and open new doors for previously untargetable cell types.

RESULTS

Vector design and approach

To date, most surveys of AAV tropism in mice have examined only a few serotypes focusing on small subsets of possible tissues, using readouts that lack cell type specific resolution. To comprehensively evaluate AAV tropism in mice, we designed an AAV vector with a bi-directional chicken beta actin (CB) promoter with a cytomegalovirus (CMV) enhancer element expressing both ZsGreen and Cre recombinase (Figure 1A).³² Recent work indicates that the Cre-mediated activation of a genomically encoded fluorophore is a more sensitive method of detecting AAV transduction than AAV-mediated delivery.^{33,34} We chose to use the Ai9 reporter mouse strain, which harbors a CAG promoter-driven tdTomato bright red fluorescent

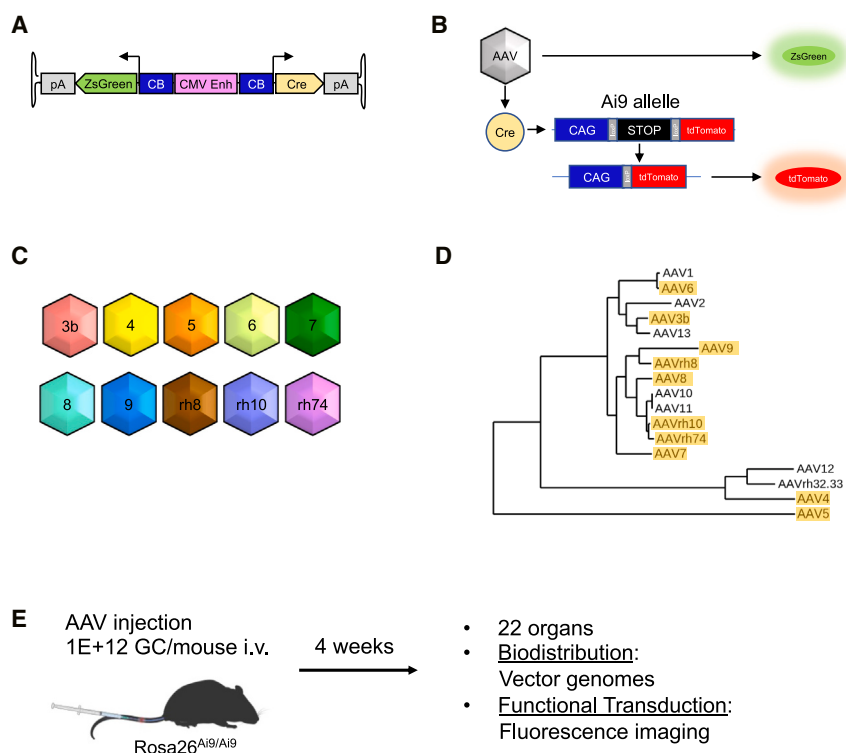


Figure 1. Experimental design for the analysis of AAV tropism survey in mice

(A) AAV vector design. CMV enhancer element/bidirectional CB promoter allows for expression of both Cre and ZsGreen from the same AAV vector. (B) Detection. Cells transduced with AAV will express both the ZsGreen fluorescent reporter and Cre recombinase to activate the Ai9 allele. ZsGreen fluorescence reports levels of active AAV transgene expression, while TdTomato fluorescence permanently marks cells with a history of transduction. (C) Serotypes. The transgene cassette was packaged into the indicated capsid serotypes. (D) Phylogenetic comparison of AAV serotypes. VP1 capsid protein sequences relationships between different natural AAV serotypes. Serotypes used in this study are highlighted in orange. (E) Experimental workflow. For each serotype, two cohorts of four to six male and four to six female mice were injected intravenously and dissected 4 weeks later. From one cohort, genomic DNA was purified from a panel of organs for qPCR analysis. From the second cohort, the same organs were prepared and imaged by fluorescent microscopy.

reporter gene at the *Rosa26* locus, with an intervening *loxP*-STOP-*loxP* cassette.³⁵ The STOP cassette can be removed by Cre, permanently activating tdTomato expression. The Ai9 allele was carried on the C57BL/6J genetic background (a strain that expresses both AAVR as well as the LY6A receptor). We reasoned that the delivery of this bidirectional transgene into Ai9 mice would allow us to simultaneously track cells with active transgene expression (ZsGreen), and those cells with a history of Cre expression (tdTomato) (Figure 1B). ZsGreen levels are expected to vary as a function of AAV copy number per cell, while tdTomato provides a binary readout of transduction with a particularly bright and uniform signal, also marking daughter cells. Additionally, biodistribution of the AAV genome can be measured for a quantitative readout of tissue uptake in relation to transgene expression.

AAV serotypes were chosen to reflect a diversity of capsid sequences (Figure 1C). As secondary criteria, we restricted our choices to natural serotypes of highest preclinical and/or clinical relevance. Our final list included: AAV3b, AAV4, AAV5, AAV6, AAV7, AAV8, AAV9, AAVrh8, AAVrh10, and AAVrh74 (Figure 1C).^{17,36–40} Figure 1D and Table S1 illustrate the relationships between capsid amino acid sequences. Our experimental workflow is outlined in Figure 1E. Five-week-old mice, homozygous for the Ai9 allele at the *Rosa26* locus, were injected intravenously through the tail vein. This route of administration was chosen to allow even distribution throughout the body. The AAV dosage was 1E12 genome copies (GC) per mouse, equivalent to approximately 5E13 GC/kg, intended to maximize the ability to detect extra-hepatic transduc-

tion. To reveal sexual dimorphism with respect to AAV tropism, and provide sufficient power for statistical analysis, four to six mice of each sex were injected. Four weeks after injection, a panel of 22 organs was harvested and prepared for either DNA isolation or fluorescent imaging (Table S2).

Biodistribution

Total DNA was prepared from each organ sample and assayed for the viral GC number by qPCR. Heatmap summaries of the results are presented in Figure 2A, while the same data for a subset of individual serotypes are shown in Figure 2B and Table S3. As expected for an intravenous injection, most serotypes (AAV5–9, rh8, rh10, and rh74) displayed the strongest tropism for liver in both male and female mice, with values exceeding 10^6 GC/ μ g of genomic DNA. Significant levels of AAV vector genomes were also observed in samples from adrenal glands, gastrocnemius muscle, diaphragm, heart, and skin. Two serotypes were notable exceptions to this pattern. AAV3B was found to have low transduction overall, with a total combined delivery to all organs of approximately 10^5 GC/ μ g DNA and did not target the liver. AAV4 displayed tropism for the lung (1.3×10^6 GC/ μ g DNA for males and 2.3×10^6 GC/ μ g DNA for females) and did not seem to transduce the liver effectively. These results closely match those found previously.³¹

This comprehensive survey of serotypes allows for quantitative comparisons of delivery within a single organ and across sexes. Figure S1 and Table S4 identify the serotypes that produced the highest vector GC number in each organ. For example, in lung, AAV4 produced the highest viral GC number in both male and female mice. In a similar manner, AAV8 was the most effective serotype for transducing

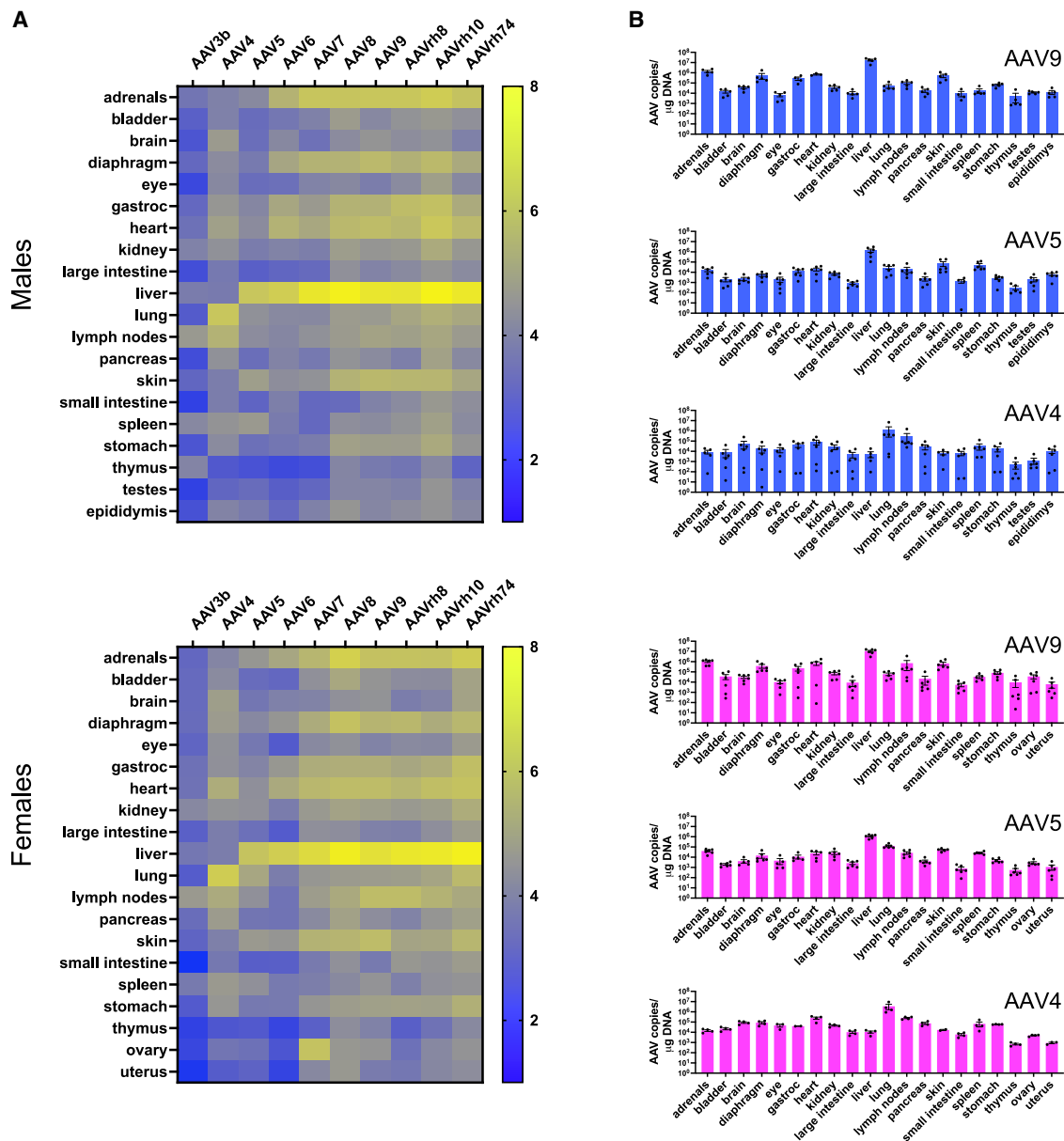


Figure 2. Biodistribution of 10 AAV serotypes across 22 different tissues

(A) Viral biodistribution summary. Total DNA was isolated from each organ in individual mice. Absolute copy numbers of viral genomes were measured by qPCR, and calculated as AAV GCs per microgram of input DNA and shown on a log₁₀ scale. $n = 4-6$ for each serotype and each sex. (B) Sample tissue comparison. qPCR results are plotted to show all organs assayed for a particular serotype. Results from male (blue) and female (pink) mice injected with AAV9, AAV5, and AAV4 are shown, with somatic organs listed in alphabetical order, and reproductive organs at the end. Error bars represent $\pm$ SEM.

bladder. Interestingly, there was some discordance between male and female mice with respect to the best performing serotypes. In male mice, AAVrh10 produced the highest combined viral GC number (5.1×10^7 GC/ μ g DNA), as well as being the top scorer for fourteen of 20 organs. In female mice, AAVrh74 produced the highest combined viral load (5.1×10^7 GC/ μ g DNA) and was the top scorer in 9 of 20 organs. Overall, some differences were observed between

male and female mice (Figure S2), including more efficient transduction of liver in males versus female mice, as previously noted.^{41,42}

Functional transduction

The ability of an AAV to mediate functional transduction resulting in fluorescent protein expression was assessed by imaging. As mentioned above, we used two simultaneous strategies to produce

fluorescent reporter molecules: direct fluorescence by expression of ZsGreen from the vector and Cre-mediated activation of tdTomato from the Ai9 allele. Initial examination of the images revealed that significantly more cells expressed Cre-activated tdTomato than ZsGreen. This phenomenon was consistent through all serotypes and all tissues. A sample of these images comparing ZsGreen and tdTomato expression in tissues transduced with AAV4 and AAV9 is shown in Figure S3. Depending on the tissue, ZsGreen fluorescence sometimes displayed a punctate intracellular pattern, and there was generally higher background observed in the green channel. Given these factors, we restricted subsequent analyses to the tdTomato reporter.

Representative images from organs with an average high (liver), medium (adrenal glands), and low (thymus) transduction efficiency across serotypes are shown for all 10 serotypes in Figure 3. Images for all 10 AAV serotypes across the 22 different organs/tissues are shown in Figure S4. As expected from the biodistribution data, a strong tdTomato signal was observed in the liver for most serotypes with the exceptions of AAV3B and AAV4 (Figure 3A). This result provides an initial demonstration of the concordance between viral GC number and functional transduction measured by a fluorescent reporter. Likewise, tdTomato fluorescence images from the adrenal glands matched the biodistribution data. In this case, fluorescent imaging revealed a consistent pattern of tdTomato expression in the cortex, especially toward the center of the organ, while the medulla was relatively untransduced (Figure 3B). This result demonstrates the advantage of using fluorescent imaging to detect transduction: the ability to identify a particular region or cell type with a different AAV tropism within the intact organ. Finally, despite the low viral GC in several tissues for a variety of serotypes, Cre-mediated tdTomato expression was observed in a handful of cells, for example in the thymus with AAV6 and AAV7 (Figure 3C). Likewise, although fluorescent imaging revealed transduction in both testes and ovaries, the activated cells appeared somatic based on morphology and location (Figure 4). In testes, tdTomato-positive cells for all serotypes appeared exclusively outside the seminiferous tubules (i.e., within the interstitial space), as defined by DAPI staining (Figure 4A). In the ovary, tdTomato-positive cells appeared solely outside the follicles (i.e., within the medulla) (Figure 4B). Therefore, our imaging did not identify transduction of germ cells by any serotype. Hence, our technique allows the detection of rare transduction events in single cells, which may nonetheless be biologically relevant, particularly for immune cells and the germ line.

To quantify functional transduction, we developed a novel image scanning technique that quantifies tdTomato⁺ red fluorescence per total area. Our results are summarized in Figure 5 and Table S5. Notably, these results seem to closely match the viral biodistribution data from Figure 2. For both males and females, liver is the most heavily transduced organ for serotypes for AAV5–9, rh8, rh10, and rh74, reaching the maximum value of 100%. Adrenal gland, heart, and muscle (diaphragm and gastrocnemius) also produced strong signal for these serotypes. For AAV4, lung had the highest percentage of tdTomato-positive cells in males, as expected from our bio-

distribution data. For females transduced with AAV4, heart and diaphragm showed a slightly higher percentage of tdTomato-positive cell and fiber area than lung (Figures 5 and S5).

Nonetheless, some differences between the viral biodistribution data and functional transduction were observed. For example, AAVrh8, rh10, and rh74 produced a significant number of tdTomato-positive cells in the uterus (Figures 5A, S4, and S5), despite a relatively low viral GC number (Figure 2A). Therefore, we sought to establish an overall correlation for all tissues and serotypes between viral GC number and tdTomato-positive cells (Figure S6). The results indicate a strong overall correlation between viral GC number and tdTomato⁺ cells. Interestingly, the data produced sigmoidal curves. At the low end, this result suggests that a certain threshold of viral particles can enter a tissue before tdTomato allele activation by Cre occurs. At the high end, the curve flattens once tdTomato-positive cells reach 100%, and further transduction cannot be registered. Graphs with the same correlation data for individual organs are shown in Figure S7. For most tissues, a direct relationship between viral GCs and tdTomato-positive cells is evident. However, three organs—spleen, thymus, and lymph nodes—seem to show reasonably high vector GC number without a corresponding amount of tdTomato-positive cells. This apparent non-functional transduction (DNA delivery without transgene expression) may be related to the immune function of these organs or reflect promoter-specific silencing in these tissues.

AAV4 has a pan-endothelial tropism and transduces pancreatic islets

The tropism of AAV4 differed significantly from all the other serotypes tested, with poor liver transduction and a very strong signal in lung. This less studied AAV serotype is one of the most distinct in terms of capsid sequence and has a unique preference for α 2-3 O-linked sialic acid.²⁰ AAV4 was reported to exhibit a cardiopulmonary tropism based on vector genomes and luciferase imaging,⁴³ but specific cell types were not identified. To further interrogate cell type specificity, we generated AAV4 vectors expressing Cre recombinase, driven by either the CB promoter with CMV enhancer, or the EF1 α promoter. These vectors did not express ZsGreen, allowing a broader range of fluorescent markers to be used for analysis. Intravenous delivery via the tail vein of AAV4-Cre to Ai9 mice resulted in substantial tdTomato-positive signal in lung outside the airways (Figure 6A). Flow cytometry revealed that the majority of tdTomato signal originated from endothelial cells based on CD31⁺/CD45[−] expression. Remarkably, more than 85% of lung endothelial cells were transduced by AAV4 in both male and female mice. In contrast, only approximately 20% of non-immune, non-endothelial cells in the lung appeared to be transduced by AAV4 (Figure 6B).

These striking results prompted us to ask whether AAV4 might also transduce endothelial cells in other vascular beds, a historically neglected cell type for gene therapy applications. The experiment was repeated using an AAV4 vector expressing Cre driven by the EF1 α promoter. To label the luminal endothelial surface of the vasculature, DyLight649-conjugated tomato lectin was injected

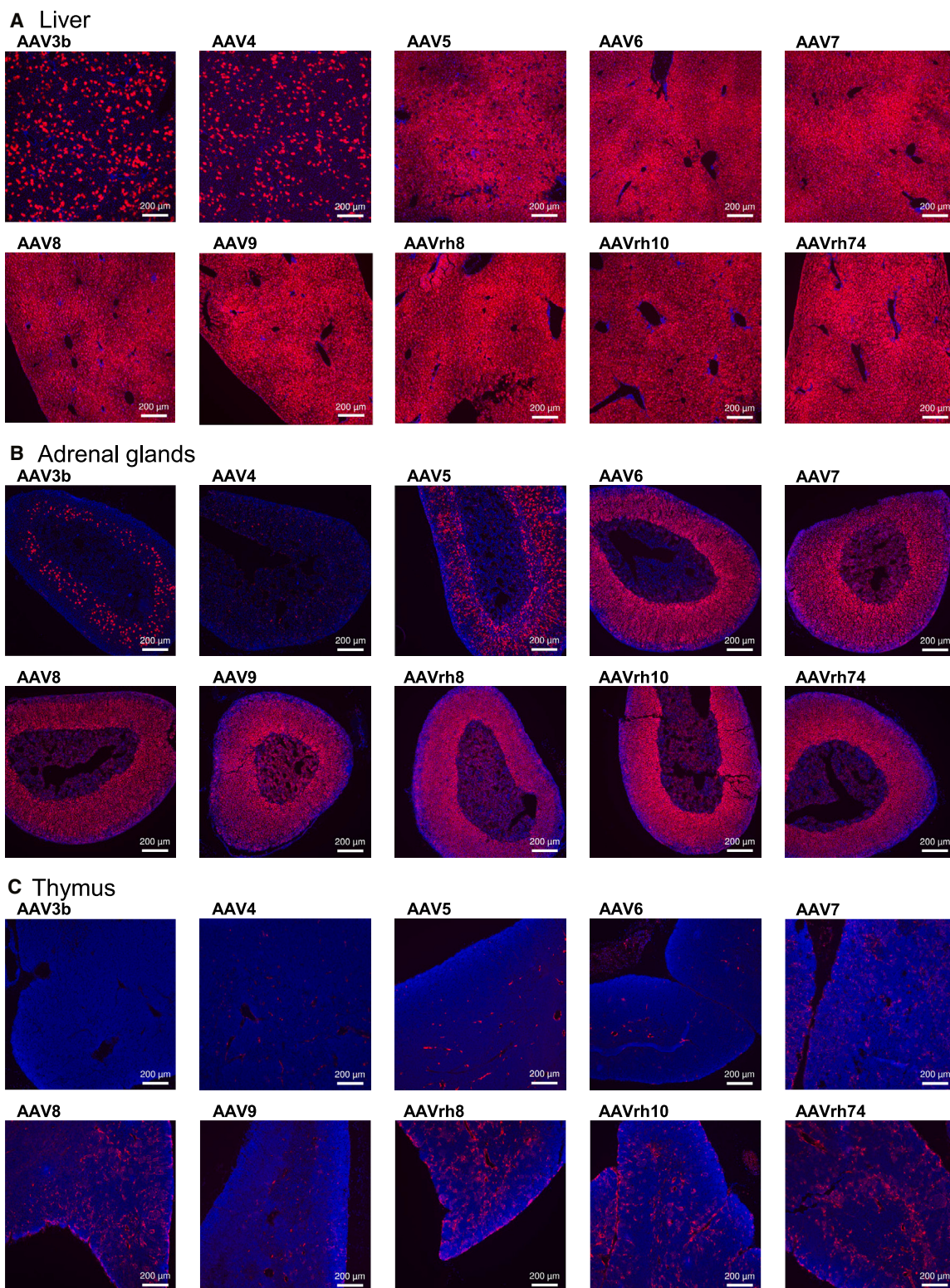


Figure 3. Functional transduction detected by fluorescent imaging

Organ samples were prepared and imaged for fluorescent tdTomato (red) indicative of functional transduction. Nuclei were stained for DAPI (blue). Representative images from (A) liver (high transduction), (B) adrenal glands (medium transduction) and (C) thymus (low transduction) for each serotype are shown. Scale bars are 200 μ m.

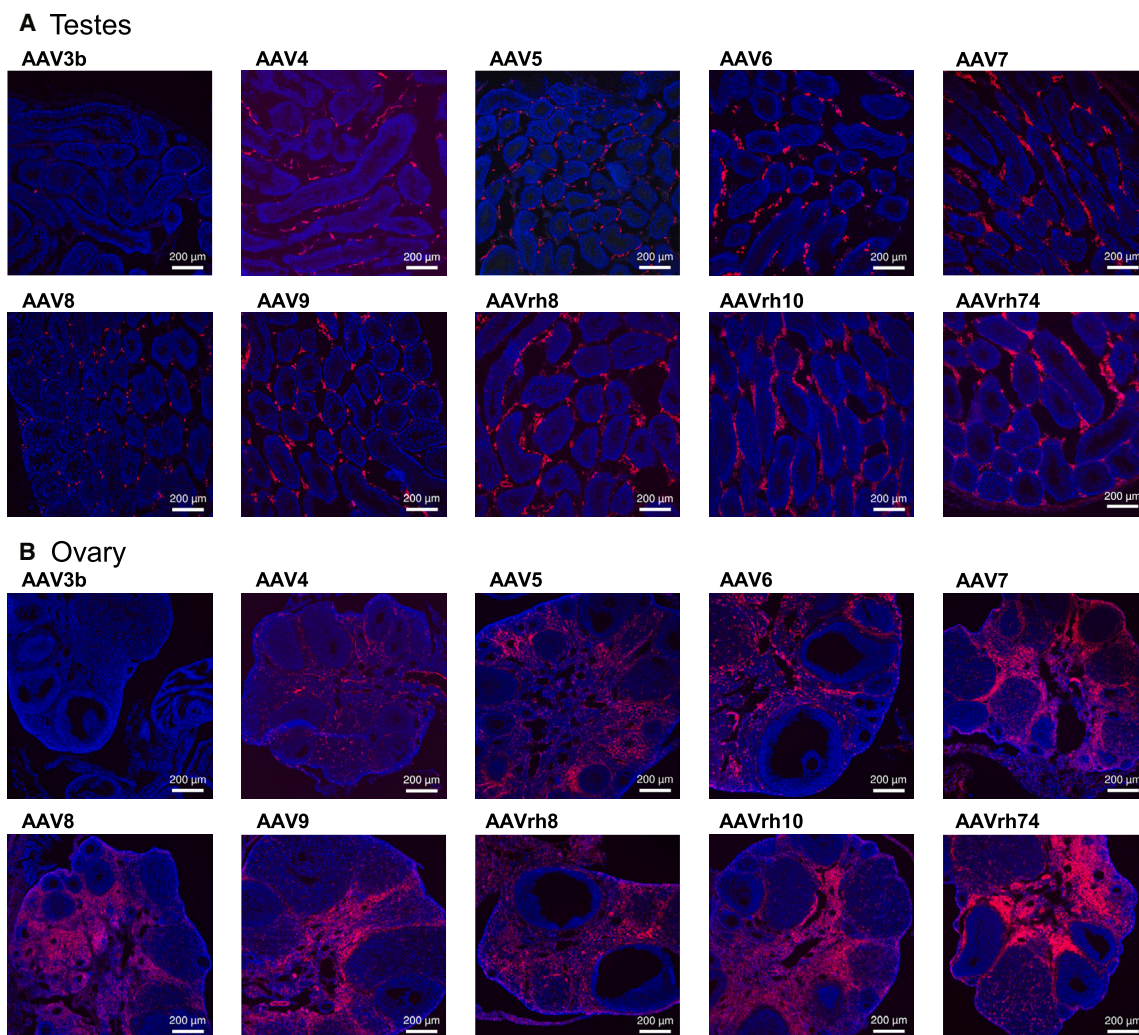


Figure 4. Functional transduction of testes and ovary

tdTomato fluorescence (red) and DAPI nuclei (blue) for tissues from mice transduced with the indicated serotypes in (A) testes and (B) ovary. Scale bars are 200 μ m.

intravenously before dissection. Sections from multiple organs were prepared at a thickness of forty microns and imaged for the fluorescent lectin and tdTomato. As expected, liver was poorly transduced with AAV4 showing only sparse tdTomato⁺ hepatocytes and possible signal in isolated liver sinusoidal endothelial cells. In stark contrast, impressive transduction of endothelial cells was seen in the stomach, bladder, adrenals, pancreas, brain, small intestine, and thymus (Figure 6C). A clear and distinct endothelial signal was also observed in the large intestine, muscle, lymph nodes, skin, ovary, and uterus (Figure S8). In separate experiments, we also confirmed strong endothelial delivery to the retina (Figure S9A). AAV4 also mediated transduction of the largest vessel in the body, hitting 26% of endothelial cells in the thoracic aorta (Figure S9B).

In addition to its apparent pan-endothelial tropism, we also noted strong signal in pancreatic islets with AAV4 (Figure S4). To confirm

this result, we performed a second experiment with the Cre-only AAV4 construct. Co-staining with an antibody to insulin confirmed efficient transduction of beta cells with AAV4 (Figure 6D). We re-examined other serotypes for transduction of pancreatic islets in our initial experiments (Figure S10A). Although AAV8 seemed to have some tropism for islets, it was much less specific, with strong tdTomato signal from surrounding acinar cells following intravenous delivery (Figure S10B). AAV5 also seemed to transduce islets, although much less efficiently than AAV4. Strong transduction of pancreatic acinar cells was seen for AAV7, 9, rh8, rh10, and rh74.

Hematopoietic and immune cell transduction

We also analyzed transduction efficiency of bone marrow-derived hematopoietic stem and progenitor cells, as well as circulating leukocytes. Cells isolated from bone marrow were stained with the appropriate antibodies to identify functional transduction by Cre-induced

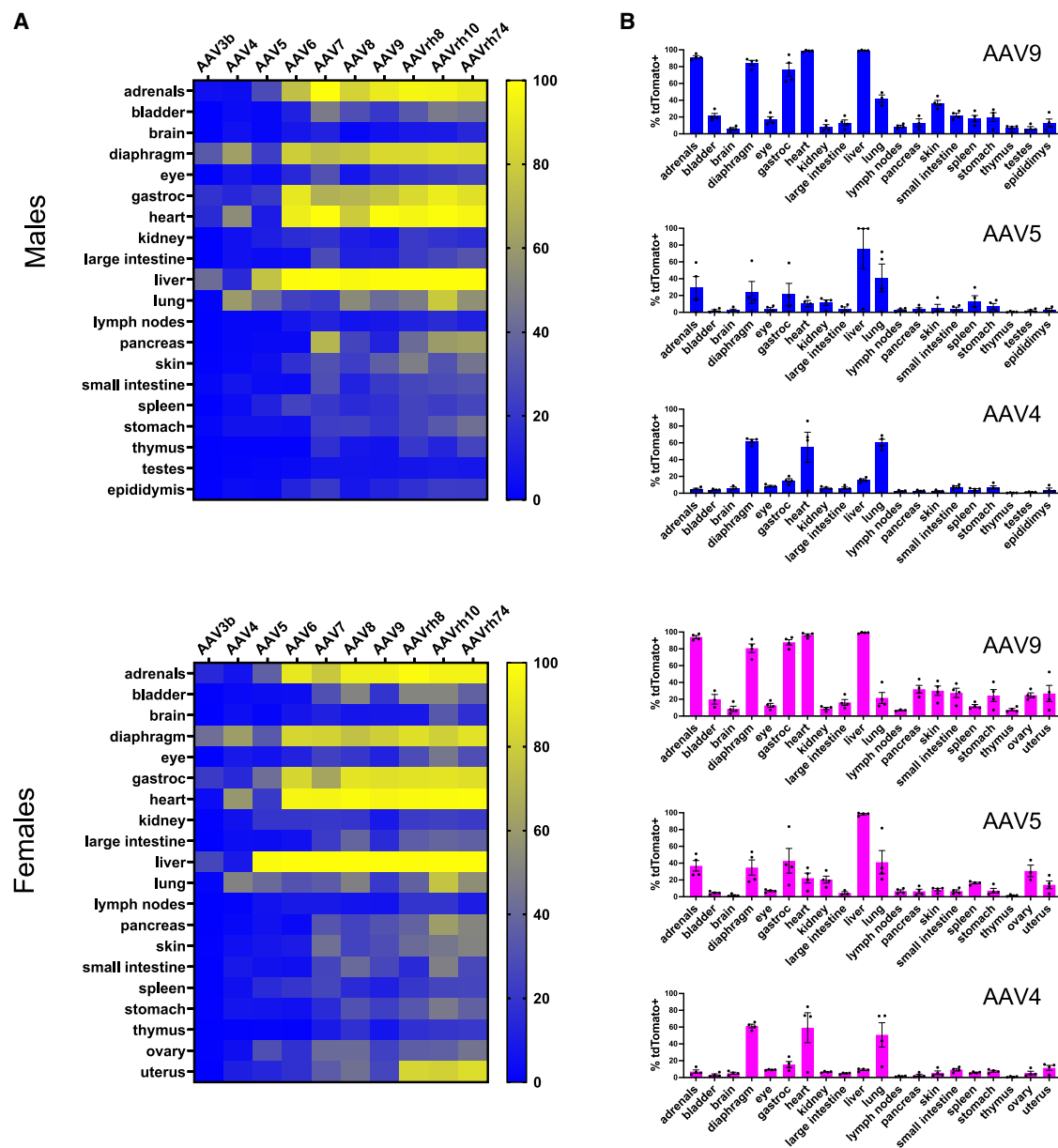


Figure 5. Quantification of functional transduction across tissues

(A) Image quantification. tdTomato signal in each image was measured to quantify transduction efficiency from four images for each organ, serotype and sex, and expressed as a percentage of maximum coverage ($n = 4$). Results are shown as a heatmap. (B) Sample tissue comparison. Image quantification results are plotted to show all organs assayed for a particular serotype. Results from male (blue) and female (pink) mice injected with AAV9, AAV5, and AAV4 are shown, with somatic organs listed in alphabetical order, and reproductive organs at the end. Error bars represent $\pm$ SEM.

tdTomato expression by flow cytometry (Figures 7 and S11; Table S6). Strong transduction of short- and long-term HSCs and multipotent progenitor cells (MPP1, MPP2, and MPP3) was observed for multiple serotypes. For AAVrh8, rh10, and rh74 transduction efficiency was greater than 50% in both males and females (Figures 7A, 7B, and S11A). Interestingly, AAV7 seemed to transduce HSCs more effi-

ciently in males than in females (66% vs. 53%; $p = 0.03$), while for AAV8, females showed a higher transduction efficiency in these cells compared with males (67% vs. 48%; $p = 0.02$). Interestingly, AAV9, which exhibited broad biodistribution and functional transduction across multiple organs, was relatively inefficient in bone marrow-derived cells.

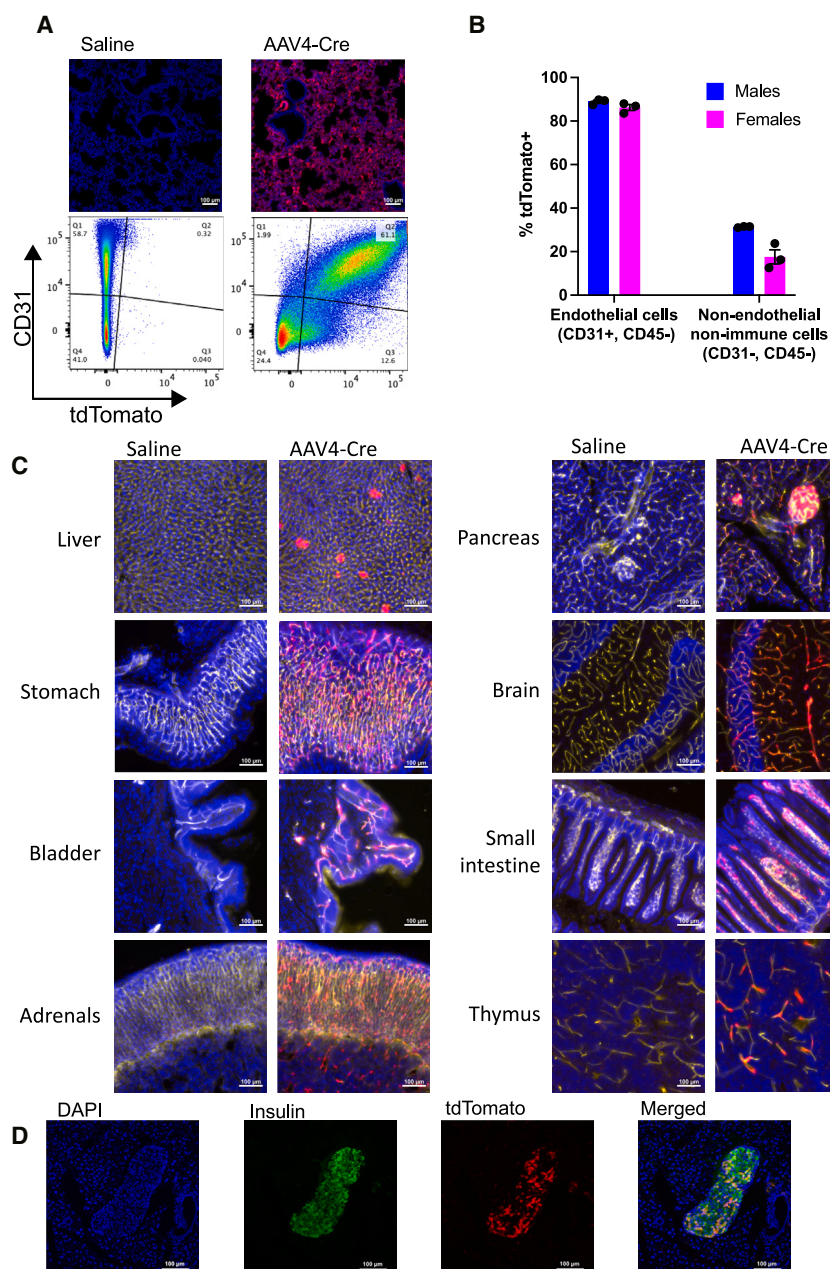


Figure 6. AAV4 exhibits a unique tropism for endothelial cells and pancreatic beta cells

(A) Transduction of pulmonary endothelial cells by AAV4 - Fluorescent images and flow cytometry plots showing induction of tdTomato expression (upper right quadrant) in endothelial cells (CD31⁺, CD45⁻) following injection with AAV4-Cre. (B) Quantification of pulmonary endothelial cell transduction flow cytometry results. Error bars represent $\pm$ SEM. (C) Transduction of endothelial cells in multiple tissues. Tissue sections from negative control and AAV4-Cre injected mice are shown, with DAPI staining for nuclei (blue), DyLight649-lectin staining for endothelial cells (yellow) and tdTomato (red). (D) Pancreas sections were stained with a fluorescently labelled insulin antibody and imaged by confocal microscopy. Scale bars are 100 μ m.

with bone marrow-derived cells. For all cell types and serotypes, mean transduction efficiencies were less than 10% (Figures 7D and S11C). Nonetheless, some differences were observed. CD11c⁺ monocytes, which includes dendritic cells, were more highly transduced than other cell types, especially by AAVrh74. In contrast, natural killer (NK) cells showed a stronger transduction by AAV5 than other serotypes. Our results have produced the most complete survey of AAV tropism in the mouse hematopoietic system to date.

DISCUSSION

Currently, the most cited survey of AAV tropism in mice comes from the work Zincarelli et al.³¹ This study explored the viral bio-distribution of nine AAV serotypes (AAV1–9) across eight organs (brain, kidney, heart, liver, lung, skeletal muscle, and testes). Systemic intravenous injection was used because it provides broad tissue distribution and is broadly therapeutically relevant. The AAVs used in the experiments carried luciferase expression cassettes under a ubiquitous promoter, allowing serial measurement of functional transduction

This analysis was extended to lineage-committed cell types within the bone marrow. Committed progenitor cells were generally less transduced than stem cells and multipotent progenitors (Figures 7C and S11B). Common myeloid progenitors were more strongly transduced than common lymphoid progenitors in both males and females by multiple AAV serotypes. However, the more differentiated myeloid progenitor cell types, granulocyte-macrophage progenitors, and megakaryocyte-erythrocyte progenitors were less transduced. Circulating leukocytes were also assayed for transduction, following isolation of peripheral blood mononucleated cells. Notably, overall transduction efficiency was significantly lower for these cells compared

through bioluminescent imaging. Their results identified AAV9 as having the widest organ distribution and highest transgene expression. However, this study was limited to male mice, and did not allow examination of transduction at the level of individual cells. More recent surveys have focused on assessing AAV tropism among cell types within a specific organ, including brain,^{44,45} and eye,^{44,45} or from a single serotype.⁴⁶

Here, we performed a comprehensive evaluation of AAV tropism in mice, the most widely used preclinical model organism for gene therapy. Our experiments expand on prior work in several important

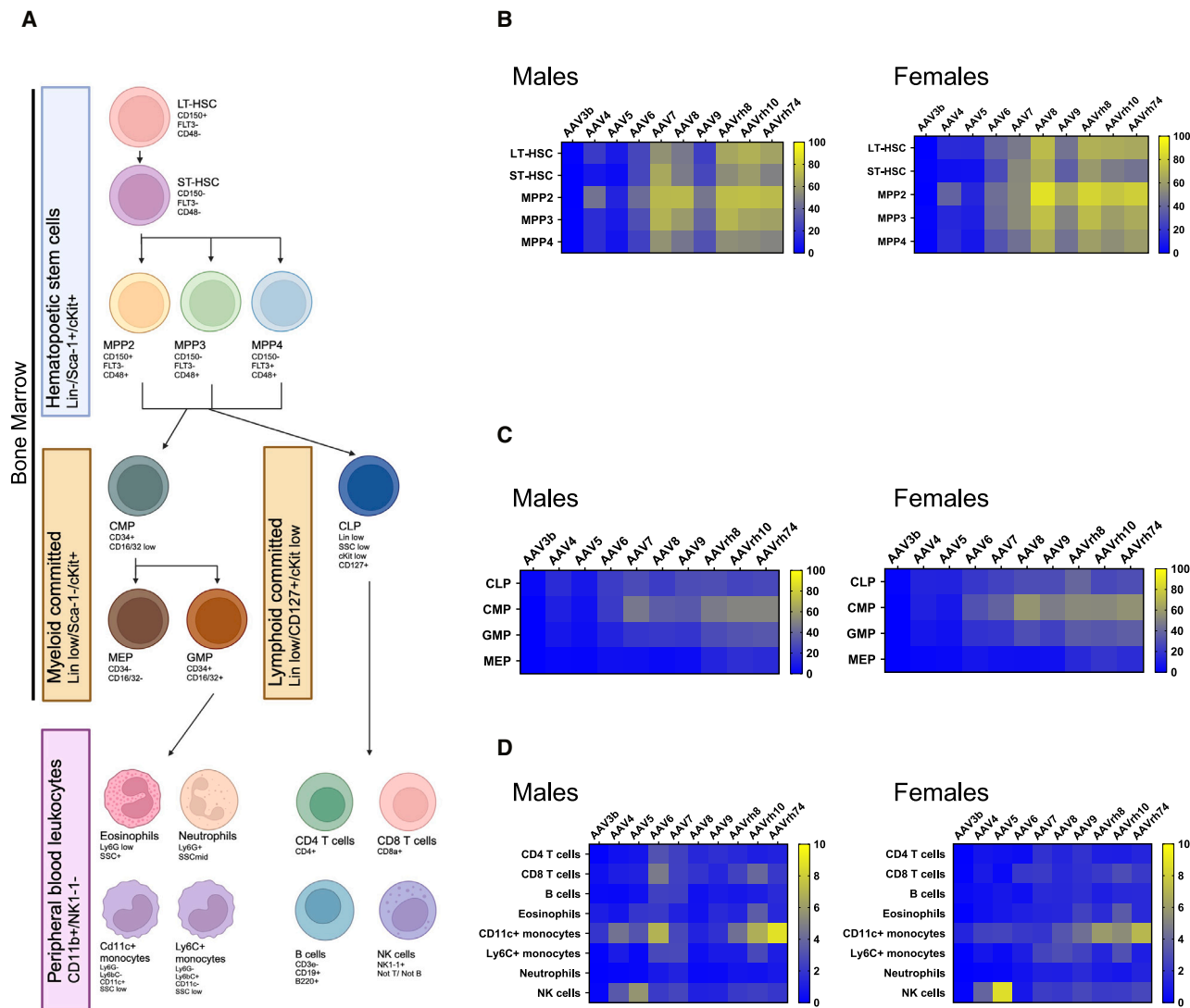


Figure 7. Transduction of HSCs, progenitor cells, and leukocytes

(A) Schematic of the HSC and leukocyte lineages tested. (B) Percentage of transduced long term, short term and multipotent progenitors from bone marrow, determined by tdTomato fluorescence via flow cytometry. Colors on heat maps indicate percent tdTomato-positive cells in each cell population. (C) Percentage of transduced myeloid and lymphoid lineage-committed cells from bone marrow. (D) Percentage of transduced circulating leukocytes (peripheral blood leukocytes) isolated from peripheral blood.

ways. We examined 10 serotypes originally isolated from humans (AAV3B, AAV5, AAV6, and AAV9) and non-human primates (AAV4, AAV7, AAV8, AAVrh8, AAVrh10, and AAVrh74), with a diversity of capsid structures. Our organ panel consisted of 22 separate tissues, as well hematopoietic stem and progenitor cells, and circulating immune cells. A dose of 1E12 GC (10-fold higher than in Zincarelli et al.) was chosen to ensure extrahepatic transduction was effectively captured. We also included both male and female mice in our analyses. Perhaps most important, we determined functional transduction by fluorescent imaging of individual organ sections. This technique allowed a detailed examination of functional transduction at the level of individual cells, rather than the broad or-

gan-level detection of transgene expression from bioluminescence or beta-gal staining that is used traditionally. Additionally, by testing serotypes individually we avoided competition for common receptors or variable representation which can confound pooled library approaches.

Results from our evaluation of viral biodistribution correlate well with the previous findings. In general, our values for viral GCs detected, calculated as GC/ μ g total genomic DNA, were concordant with previously published values.³¹ As expected, most serotypes (AAV6–9, rh8, rh10, and rh74) preferentially transduced the liver. In addition, muscle (gastrocnemius and diaphragm), as well as heart, was strongly

transduced by these serotypes. Notably, these serotypes are closely related evolutionarily, reinforcing the role of capsid structure in determining viral tropism.¹⁷ AAV5, which is the phylogenetically most distant from other serotypes, also transduced liver but failed to effectively transduce muscle and heart.

By examining both male and female mice, we were able to uncover evidence of sexual dimorphism with respect to AAV tropism across the natural serotypes tested. As expected, stronger liver transduction was seen in males than females for most serotypes.⁴¹ The notable exception was AAVrh74, which produced a pronounced preference for transduction in female mice, not only in liver, but across all organs. In contrast, AAVrh10 showed a transduction preference for males for all organs except lymph nodes, which were very weakly transduced in both sexes. AAV6 also demonstrated a bias toward transduction in males across multiple organs. For most other organs and serotypes, sex differences were relatively minor. Nonetheless, sexual dimorphism remains an important consideration for AAV-mediated gene therapy, given the potential impact on efficacy.⁴⁷ The extent and nature of sexual dimorphism in human AAV gene therapy remains an open question.

To generate data revealing functional transduction, we used a dual fluorescent reporter strategy. The transgene contained within our AAVs expressed both the ZsGreen fluorescent reporter and Cre recombinase. In the Ai9 reporter strain used in the studies, Cre expression drives activation of tdTomato fluorescent reporter expression. This event permanently marks the targeted cell and its daughter cells after transduction. During our initial examination of our fluorescent images, we noted that Cre-driven tdTomato expression was visible in significantly more cells than ZsGreen, irrespective of the tissue or serotype. tdTomato activation also had an excellent signal-to-noise ratio, allowing for more accurate discrimination of positive cells from background fluorescence. Hence, we found that this technique was the most sensitive mechanism for detecting transduction, corroborating recent findings.³³

Our results strongly reinforce the utility of fluorescent imaging for the detection of functional AAV transduction. We detected rare transduction events in individual cells from multiple tissues where the biodistribution data suggested a low tropism for a particular serotype. From a clinical point of view, rare transduction events by a therapeutic AAV may be consequential. Therefore, our results demonstrate the need for careful analysis of transduction of non-target tissues. In particular, we focused on sex organs for evidence of non-target transduction, given concerns over the possible germline transmission of genome alterations.

We noted significant transduction of both testes and ovaries. However, in testes only somatic cells located outside the tubules were transduced. Likewise in ovaries, the transduced cells appeared outside the follicles, also indicating that they were not part of the germ line. Schuettrumpf et al.⁴⁸ found that vector genomes are detectable in semen from rabbits receiving an intravenous injection of AAV2 vec-

tors. In follow-up work, Favaro et al.⁴⁹ analyzed the risk of germline transmission following intravenous delivery of AAV2 and AAV8 vectors expressing factor IX to rabbits. Although rAAV genomes were shed in rabbit semen, these were thought to be derived primarily from epithelial cells from the genitourinary tract. Studies in mice confirm that AAV genomes are shed in semen, but consistently show a lack of transmission to offspring,^{50–52} even when mouse spermatozoa are directly exposed to high doses of AAV2 before *in vitro* fertilization.⁵³ The risk of germline transmission could theoretically be greater with AAV-based genome editing approaches. Nonetheless, we did not find evidence of germ cell delivery in either male or female mice using a highly sensitive reporter that permanently marks transduced cells.

Strong physical and functional transduction by AAV6–9, plus rh8, rh10, and rh74 serotypes was also observed in adrenal glands, an organ not included in previous comparative surveys of AAV tropism, but which has previously been found to be transduced by AAV9.⁴⁶ This result validates our decision to include smaller, less frequently examined organs in our survey. Imaging revealed that the transduction occurred primarily in the cortex, rather than the medulla. Adrenal glands should hence be considered not only as a tissue potentially targetable with AAV delivery, but also one to consider in terms of side effects originating from non-target tissues. Further work is needed to determine to what degree this organ may also be transduced in non-human primates.

Two serotypes differed in the liver-dominant tropism most frequently observed, with a viral GC number 100-fold lower in this organ compared with the other serotypes. Interestingly, these two serotypes, AAV3B and AAV4, differed in their tropism in non-hepatic tissues. AAV3B did not seem to transduce other organs in the place of liver. Instead, it did not seem to have an affinity for any specific organ. This effect may be specific for mice, since previous studies have shown that while this serotype has a poor tropism for mouse liver, it can strongly transduce hepatocytes from humans and non-human primates.²⁹ The greatest physical transduction for AAV3B was observed in lymph nodes and spleen, although fluorescent imaging did not indicate functional transduction of these organs by this serotype. It is possible, given the role of these organs in mediating immune responses, that, in these tissues, virus particles are taken up but processed to generate an immune response, rather than for gene delivery. These biodistribution data provide useful information about potential latent reservoirs of AAV that impact adaptive immunity and long-term efficacy.

A major finding of this work is the pan-endothelial tropism of AAV4, a lesser studied AAV serotype that is difficult to produce at high titer. AAV4 transduced the liver relatively poorly, but strong transduction was observed in the lung, as previously reported.^{31,43} We found that pulmonary endothelial cells are the primary cell type targeted in the lung with AAV4 by the intravenous route. We also noted substantial and highly specific endothelial cell transduction by AAV4 in many other organs (>10 different tissues), opening the door for

endothelial-directed gene therapies. AAV4 differs from other serotypes not only in capsid structure but also in its unique affinity for α 2-3 O-linked sialic acid.^{20,43} Interestingly, AAV4 is the only serotype we tested which does not require AAVR for cell entry, potentially explaining its poor liver tropism. It is also not clear why AAV4 is so effective for endothelial cells, although other serotypes (i.e., AAV8, AAV9) are presumably transcytosed through the endothelial barrier to enter parenchymal cells. Further work is needed to identify the endocytosis receptor for AAV4 and to define its intracellular trafficking itinerary, which results in endothelial transduction rather than transcytosis. In terms of pancreatic islet delivery, AAV8 has long been known to transduce beta cells following intravenous or intraperitoneal injection in mice.^{54–58} Previously, AAV8 was used to transduce acinar cells with three key beta cell transcription factors, attempting to convert acinar cells to restore depleted beta cells in diabetes.⁵⁹ However, AAV4 seems to have the greatest selectivity for beta cells over the acinar cells in pancreas with systemic intravenous delivery. Thus, our observations may be useful for the design of delivery vehicles for gene therapies that could directly target expansion of beta cells in diabetes.

There is growing interest in targeting hematopoietic stem and progenitor cells for *in vivo* gene therapy and genome editing. Previous data suggest significant transduction of bone marrow-derived HSCs by AAV6, AAV8, and AAV9.⁶⁰ Our data corroborate and extend these results. AAV-based gene editing of such cells within the bone marrow niche would allow for life changing gene therapies. For example, *ex vivo* gene editing therapies have just been approved for sickle cell disease and beta thalassemia. However, the high cost of such therapies combined with the difficult logistics of bone marrow conditioning means that they are unlikely to be fully translated on a global scale to the populations who would most benefit. A single-dose AAV gene therapy that can target the bone marrow following systemic administration would be very powerful. It is also noteworthy that some immune cell populations were transduced in peripheral blood, primarily CD11c⁺ monocytes, NK cells, and CD8 T cells, albeit with much lower efficiency. These cell types are potential targets for cancer immunotherapies (i.e., chimeric antigen receptor T and chimeric antigen receptor NK cells), as well as gene editing to eradicate residual HIV infection. Effectiveness of such therapies will depend both on the mode of action, as well as efficiency of delivery. This also suggests that unintended transduction of immune cell populations may play an underappreciated role in the safety and efficacy of AAV gene therapies in general.

The natural AAV capsids examined in this paper are amenable to engineering by multiple means, including rational design,⁶¹ PCR shuffling,⁶² mutagenic PCR,⁴³ library screens,⁶³ peptide display,⁶⁴ biopanning,⁶⁵ and directed evolution.^{66,67} PCR shuffling has been a very successful approach, taking advantage of sequences already determined by evolution to be compatible with capsid assembly. Creating capsids rearranged from the naturally occurring serotypes can confer new and unexpected properties. For example, AAV-DJ is a chimeric capsid of AAV2/AAV8/AAV9 with enhanced transduc-

tion of liver and lower extrahepatic transduction, along with a reduced innate immune response.⁶² Likewise, AAV-LK03⁶⁸ and AAV-NP59⁶⁹ are shuffled capsids selected based on their improved affinity for human over murine hepatocytes. Peptide insertion mutants, most often into the AAV2 or AAV9 capsid, can have a profound effect on vector tropism. For example, natural AAV serotypes are resistant to crossing the blood-brain barrier. Screening of seven-amino acid peptide sequences inserted into the AAV9 capsid revealed a variant, named AAV-PHP.B, that was 40-fold more efficient at transducing the CNS.⁶⁴ Additional peptide screening resulted in the identification of two further artificial capsids: AAV-PHP.eB, with enhanced transduction of the brain, and AAV-PHP.S, with enhanced transduction in dorsal root ganglion and enteric and cardiac neurons.⁷⁰ Other notable peptide insertion mutants into AAV9 have been engineered for improved muscle delivery, such as AAVMYO⁷¹ and MyoAAV,⁷² among others. There have also been considerable efforts to define the capsid regions involved in receptor binding of many natural serotypes. Purposeful mutagenesis of these residues also can afford major gains, by detargeting particular organs such as liver.^{73,74} We anticipate that our data on the 10 natural serotypes in this paper will support further capsid engineering efforts.

We administered AAVs by intravenous injection to maximize tissue distribution, avoiding any physical restriction to any particular organ. However, it is well recognized that the route of administration has a tremendous impact on tropism. For example, the natural serotypes tested in our studies do not cross the blood-brain barrier efficiently, but could be effective at transduction if delivered to the correct biological compartment. A previous study using direct intracerebral injection of five natural AAV serotypes revealed significant transduction for AAV5, AAV7, and AAV9, with reduced tropism for AAV2 and AAV8.⁷⁵ These results differ from our biodistribution and functional transduction results, and almost certainly reflect the difference in delivery route. A recent survey of 20 natural and engineered AAV serotypes delivered to mouse lung by intratracheal instillation identified a transduction efficiency of 30% for AAV4 in airway epithelium, but insignificant transduction in pulmonary endothelial cells. These results reinforce the critical role that delivery route plays in determining AAV tropism.⁷⁶

The experiments described in this paper were limited to Ai9 mice on a C57BL/6J genetic background. We chose this strain because of its wide use in research, particularly in the context of preclinical gene therapy. Nonetheless, differences in AAV tropism among mouse strains have been noted, and variances in transduction patterns may occur. For example, intracerebral delivery of five natural AAV serotypes (AAV2, AAV5, AAV7, AAV8, and AAV9) revealed significant differences between C57BL/6 mice and FVB mice.⁷⁵ AAV5 and AAV7 produced higher transduction in the cortex of C57BL/6 mice than FVB mice. Contralateral spreading of transduction was also markedly reduced in FVB compared with C57BL/6 following intrastriatal injection. In another example, the LY6A receptor is required for blood-brain barrier transcytosis and CNS transduction of the

engineered AAV9 derivative AAV-PHP.B following intravenous administration.³⁰ C57BL6/J and FVB strains both express the LY6A receptor, but other laboratory strains such as BALB/c and CB do not, making them nonpermissive to CNS delivery via the intravenous route. Similarly, it is likely that AAVR expression also varies across strains which may result in differences in overall or tissue-specific transduction. Genetic manipulation of endogenous AAVR expression has been used to improve surveys of hepatocyte tropism in human liver chimeric mice,⁷⁷ as well as to hardwire artificial AAV tropism for basic science research in a Cre-dependent manner.⁷⁸ Systematic examinations of AAV tropism across inbred mouse strains could be combined with genetic screens to learn more about AAV biology, but is beyond the scope of the current work.

It remains to be determined how translatable these tropism data are to humans. Although there are many examples where tropism does not translate well across species, many common themes exist. For example, most approved AAV gene therapy products involve AAV2, AAV5, AAV8, or AAV9 capsids. These vectors were tested extensively in mouse, primate, and/or canine models before progressing to human clinical trials. Vectors capable of transducing liver, skeletal muscle, CNS, and retina are generally effective across multiple species, albeit with varying efficiency. An important endeavor will be comparing tropism data from mice reported here, with that obtained in non-human primates, as well as with clinical outcomes. Ideally, cross-species-compatible vectors can be developed for specific tissues using these data as a starting point, thus streamlining the development of gene therapies from initial proof of concept in mice to ultimate application in humans.

All data from these experiments are available through the Somatic Cell Genome Editing Consortium Toolkit (<https://scge.mcw.edu/toolkit>) for further exploration and analysis. We believe this resource will be immediately useful for other researchers engaged in capsid engineering to target specific cell types or tissues in mouse models, as well as for avoiding unintended delivery to non-target tissues. These data will inform the community of the most effective vectors for pre-clinical gene therapy in mice, and more broadly improve our understanding of AAV biology. Moreover, the detailed documentation of targeted tissues and cell types, including the germline, is critical to the ongoing development and deployment of genome editor-based therapeutics.

MATERIALS AND METHODS

rAAV production

AAV vectors were produced by standard triple-transfection in HEK293 cells. Three days post transfection, cells and culture media were harvested, and crude lysates were prepared. Downstream purification was carried out by two rounds of CsCl sedimentation followed by dialysis. Purified AAV vectors were sterilized using a 0.22-micron filter and formulated in PBS with 5% sorbitol and 0.001% of F-68. AAV vector genome titers were determined by droplet digital PCR. AAV vector purity was assessed by gel electrophoresis followed by silver staining.

AAV transduction in mice

Mice used in experiments were from the Ai9 strain, B6.Cg-Gt(ROSA)26Sor^{tm9(CAG-tdTomato)Hze}/J (Madisen et al. 2010, JAX# 007909). At 5 weeks of age, male and female mice were injected intravenously through the tail vein with rAAV. Injections were carried out in parallel at BCM and JAX using the same viral batch. Four to six mice of each sex were injected with AAVs of each serotype. Dosage was 1E12 GC/mouse, in a total volume of 0.2 mL PBS. Following injection, mice were housed under standard conditions, and fed *ad libitum*, while monitoring for weight loss or other health issues. After 4 weeks, the mice were humanely euthanized and dissected for biodistribution and functional transduction analyses. In experiments involving labeling of endothelial cells, 100 mg of DyLight649-labelled *Lycopersicon esculentum* (tomato) lectin (Vector Laboratories, DL-1178-1) was injected intravenously through the tail vein in a volume of 0.1 mL just before euthanasia and dissection. Tissue samples from each organ were prepared for imaging as described below. In addition, small samples (approximately 50 mg) were flash frozen in liquid nitrogen and stored at -80°C for subsequent genomic DNA preparation. For brain samples, one hemisphere cut sagittally was used for DNA preparation, to account for possible differences in AAV transduction between regions. All animal work was performed in accordance with the guidelines for animal care at Baylor College of Medicine (BCM) and the Jackson Laboratory (JAX), under approved Institutional Animal Care and Use Committee protocols.

DNA isolation and qPCR

Total DNA (viral and mouse genomic) isolation from frozen samples was performed using DNeasy Blood & Tissue kits (Qiagen) according to the manufacturer's instructions. Tissue disruption steps were repeated twice on muscle and heart samples to ensure quantitative recovery of DNA. The DNA concentration in each sample was assayed by fluorescence using a Quant-iT Broad Range kit (Invitrogen). Absolute viral GC numbers in each sample were determined by SYBR-based qPCR using a standard curve with the AAV transgene plasmid and validated primers specific to the Cre recombinase transgene (forward: 5'-TGA CGG TGG GAG AAT GTT AAT C-3'; reverse: 5'-GCT ACA CCA GAG ACG GAA ATC-3').³³ Results were calculated as AAV GCs/ μg total input DNA.

Fluorescent imaging

Tissue preparation and imaging at JAX were carried out as follows. Twelve mice per AAV serotype treatment group were euthanized via CO₂ asphyxiation. For collection of blood cells and bone marrow for flow cytometry analysis, immediately following confirmation of expiration, peripheral blood (250–500 μL) was collected via cardiac puncture into K2-EDTA microtainer tubes (BD #365974) and leg bones (either 2 two or one femur and one tibia) were collected into cold $1\times$ PBS in 12-well plates for isolation of bone marrow. The remaining tissues: brain (sagittal bisection), eyes, inguinal lymph node, skin, bladder, ovary, uterus, epididymis, testes, large intestine, gastrocnemius muscle, diaphragm, heart, thymus, lung, spleen, kidney, liver, adrenal gland, pancreas, small intestine, and stomach were collected into seven vials and immersion fixed in 25 mL, cold

4% PFA in 1× PBS per vial at 4°C on a shaking incubator for 4–5 h. Tissues were washed in cold 1× PBS twice and then transferred to into cold 30% sucrose/1× PBS solution on a shaking incubator overnight to equilibrate. Excess connective tissue was trimmed and tissues were combined into seven groups per block to minimize the number of blocks per mouse, and then immersed in Tissue-Tek O.C.T in cryomolds, frozen on dry ice, and stored at –80°C. Tissues were cryosectioned at 15 µm and sections mounted on Superfrost Plus slides (Thermo Fisher Scientific), allowed to air dry for approximately 10 min at room temp, and washed with 1× PBS twice for 5 min. Slides were then stained with 1:10,000 Hoechst in 1× PBS for 5 min, washed 2× with 1× PBS and slides mounted using ProLong Diamond Antifade Mountant, cured overnight, sealed with clear nail polish. Tissues were imaged using a widefield DMi8 microscope (Leica) using a 10× or 20× objective. Representative sections from each tissue were analyzed and at least three animals per sex per treatment were assessed, with most including four mice per sex per treatment. Quantification of tdTomato (RFP) signal and thus viral transduction was performed as follows: images were opened in QuPath,⁷⁹ and cells were detected using the built-in watershed cell detection function based on DAPI/nuclear signal. All images from the same tissue type were processed with the same cell detection settings, with only slight changes in parameters used for different tissues to account for changing cell and nuclear diameters (i.e., adjusting sigma microns [range, 0.5–0.9] and cell expansion microns). Background signal for RFP expression was determined empirically from uninjected control tissues from Ai9 mice imaged at defined increasing exposure times and used to set detection threshold minimums for each image. RFP signal above background threshold was scored per cell (co-localization). Data recorded per image includes the total number of DAPI nuclei (cells) and the number of nuclei, which had RFP signal above threshold (transduced cells) and we report the calculated percent of transduced RFP⁺ cells out of the total number of cells per image (RFP⁺ cells/total cells × 100%).

Tissue preparation and imaging at BCM were carried out as follows. Following euthanasia, animals were perfused briefly with PBS through a cardiac puncture, then organs were dissected. Tissue samples from each organ were placed in 8 mL of freshly prepared 4% buffered paraformaldehyde and fixed overnight at 4°C with mild shaking. The next day, fixed samples were transferred to a 15% buffered sucrose solution, then 3 h later, transferred to a 30% buffered sucrose solution for equilibration overnight at 4°C. Next, the fixed and cryopreserved samples were frozen in OCT, sectioned at 14 µm, DAPI-stained for nuclei, and imaged on an Axioscan Z.1 slide scanner (Zeiss) under a 20× objective, with the appropriate fluorescence settings, at the RNA *In Situ* Hybridization Core at BCM. Sectioning was performed to maximize the imaged tissue surface area for each organ. Three non-consecutive sections from each sample were imaged. For experiments involving lectin labeling of the vasculature, 40-µm sections were prepared to generate a wider depth of field.

Aortas were prepared for imaging by cutting the left and right carotid arteries and the left subclavian artery to release the aortic arch. Then,

the aorta was cut at the level of the diaphragm. The heart and aortic arch were removed, and the remaining thoracic aorta was *en face* opened and pinned on a dissecting dish with the luminal side facing up. The thoracic aorta was fixed overnight with 4% PFA (in PBS), then incubated overnight with 30% sucrose (in PBS) and stored in PBS at 4°C afterward. An approximately 4-mm piece of aorta was cut and stained with DAPI (1:5,000 dilution in PBS, D3571, Invitrogen) for 5 min, followed by a wash in PBS for 3 min. Then, the sample was placed flat on a glass slide with the lumen faced up and covered with a coverslip with glycerol. Direct tdTomato, DyLight649-lectin and DAPI fluorescence were detected using a Nikon A1-Rs confocal microscope using a 60× objective at the Integrated Microscopy Core (BCM). Confocal sections were acquired every 2 µm from the luminal side to the first layer of smooth muscle cells.

For immunofluorescence images of pancreatic islets, PFA-fixed and frozen pancreas sections were brought to room temperature for 5 min and then placed 0.1% Triton X-100 in PBS for 5 min. A 1-h block was performed with 5% donkey serum in PBS, followed by a 2-h incubation with primary antibodies to insulin (1:1,500; rabbit; Cell Signaling #3014s; RRID: AB_2126503) or amylase (1:1,000, mouse, Santa Cruz #sc-46657, RRID: AB_626668). Slides were washed twice in 0.01% Triton X-100 in PBS for 5 min each. Slides were then incubated with secondary antibodies for donkey anti-rabbit (1:1000; Jackson ImmunoResearch #711-175-152, RRID: AB_2340607) and DAPI (1:2,000; Sigma #32670) followed by two washes in 0.01% Triton X-100 in PBS for 5 min each and coverslip mounting with Prolong Gold Antifade (Invitrogen #P36930). Omission of the primary antibody was used as a control. Images were captured on a Nikon A1R confocal microscope.

Flow cytometry

To assess pulmonary endothelial cell transduction, three male and three female Ai9 mice were injected intravenously through the tail vein with 5E+11 GC/mouse of AAV4-CB6-PI-Cre. One male and one female mouse were injected with saline as negative controls. Mice were humanely euthanized two weeks post-injection, and lungs were excised and processed for flow cytometry using a mouse lung dissociation kit according to the manufacturer's instructions (Miltenyi Biotec). The resulting single cell suspension was resuspended in PBS/0.5%BSA/2 mM EDTA, and non-specific antibody binding was blocked with anti-mouse CD16/CD32 (BD Pharmingen #553142) at 1:100 dilution for 15 min at 4°C. Cells were stained with fixable Live/Dead Blue to exclude dead cells, and antibodies specific for the following markers: CD31 (BV605, clone 390, BioLegend #102427), CD45.2 (fluorescein isothiocyanate, clone 104, BioLegend #109806), CD11b (PerCP-eFluor710, clone M1/70, Invitrogen #46-0112-80) at a 1:100 dilution for 45 min at 4°C. Flow cytometry was performed on an LSRII (BD Biosciences) instrument, and data analyzed using FloJo software (v10, BD Biosciences).

To assess cell transduction in hematopoietic lineages, whole blood and bone marrow were collected from Ai9 mice injected with 1E+12 GC/mouse. Leg bones were kept cold in 1× PBS in 12-well

plates on ice. Bones were crushed with mortar and pestle and rinsed with AutoMACS buffer (Miltenyi Cat# 130-091-222) with BSA (Miltenyi Cat#130-091-376) (final concentration of 0.5% BSA). Liquid was filtered through Falcon 32235 5-mL tubes with cell strainer. Cells were centrifuged at $500 \times g$ for 5 min and liquid decanted. Cells were resuspended at approximately 2×10^7 per mL and 1×10^6 cells were stained with either antibody cocktail BM-A or BM-B for 30 min at 4°C. Samples were washed with PBS with 5 mM EDTA and 0.5% BSA, stained with 10 μ L of 5 μ g/mL DAPI (BD Biosciences, Cat.#564907).

Per 200 μ L of sample and analyzed on BD FACSymphony A5 with 500K live cells set as stopping gate. Whole blood for white cells (peripheral blood lymphocytes- PBL) was treated with 3 mL $1 \times$ Leinco RBC Lysis buffer for 5 min at room temperature (RT), centrifuged for 5 min at $500 \times g$, decanted, resuspended with 3 mL $1 \times$ RBC Lysis Buffer for 5 min at RT, centrifuged for 5 min at $500 \times g$, decanted, washed with 4mL PBS with 5mM EDTA and 0.5% BSA, centrifuged for 5 min at $500 \times g$, decanted, blotted on paper towel, and resuspended in a total volume of 50 μ L for staining. We added 10 μ L of PBL staining cocktail for 30 min at 4°C, washed with 2mL of PBS with 5 mM EDTA and 0.5% BSA, centrifuged for 5 min at $500 \times g$, decanted, resuspended in 200 μ L of PBS with 5 mM EDTA and 0.5% BSA, stained with 10 μ L of 5 μ g/mL DAPI, and run on the cytometer. Samples were analyzed on BD FACSymphony A5 and data analyzed using FloJo (v10, BD Biosciences).

Antibody staining panels for HSCs (HSC, BM-A), committed lymphoid and myeloid progenitors (CLP/CMP, BM-B), and peripheral blood lymphocytes are listed in Table S7. Specific cell populations were defined by the following staining characteristics: long-term HSC (CD150⁺, Flt3⁻, CD48⁻), short-term HSC (CD150⁻, Flt3⁻, CD48⁻), MPP2 (CD150⁺, Flt3⁻, CD48⁺), MPP3 (CD150, Flt3⁻, CD48⁺), MPP4 (CD150⁻, Flt3⁺, CD48⁺), CMP (CD34⁺, CD16/32 low), MEP (CD34⁻, CD16/32⁻), GMP (CD34⁺, CD16/32⁺), CD4 T cells (CD4⁺, CD3e⁺), CD8 T cells (CD8a⁺, CD3e⁺), B cells (CD19⁺, B220⁺, CD3e⁻), eosinophils (Ly6G low, SSC⁺), CD11c⁺ monocytes (Ly6G⁻, Lyb6C⁻, CD11c⁺, SSC low), Ly6C⁺ monocytes (Ly6G⁻, Lyb6C⁺, CD11c⁻, SSC low), neutrophils (Ly6G⁺, SSC mid), and NK cells (NK1-1⁺, Not T/NotB).

Statistical analysis

All possible pairwise comparisons ($k = 45$) of AAV serotypes, within sex and organ (cell type), for each type of data (biodistribution, function transduction, functional transduction by flow cytometry) were performed using the pairwise.wilcox.test() function in package: stats of R version 4.3.2. The p values were calculated by the exact method unless there were ties, in which case, the function used the chi-square approximation with continuity correction. Within each organ and sex, p values were adjusted using the false discovery method.⁸⁰

DATA AND CODE AVAILABILITY

All data and images associated with these studies are publicly available at <https://scge.mcw.edu/toolkit>, as part of the Somatic Cell Genome Editing initiative.

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AUTHOR CONTRIBUTIONS

The studies were conceived by W.R.L., J.D.H., C.J.W., D.W., G.G., C.M.L., K.J.S., and S.A.M. in consultation with O.M. and other members of the NIH Somatic Cell Genome Editing Consortium, and NIH program officials. AAV vector design and production was carried out by J.B., D.W., and G.G. Studies of biodistribution of AAV genomes were conducted by C.J.W., A.E.M., M.D.G., and M.A.C. Studies of functional transduction by fluorescence were carried out by K.J.S., E.S., J.D., Y.G., S.H., S.A.M., and C.M.L. Additional sectioning and imaging was performed by M.C.L. Analysis of endothelial cell transduction was carried out by C.J.W., M.D.G., M.A.-B., C.F.S., and M.C.L. Analysis of pancreatic islet transduction was performed by A.R.C. and M.C.L. with support from S.M.H. Expertise on confocal microscopy imaging provided by C.W.H. Generation of experimental reporter mice was achieved by D.G.L. and J.D.H. Statistical analyses of transduction data was performed by C.J.W. and S.G.H. The manuscript was written by C.J.W., W.R.L., J.D.H., K.J.S., S.A.M., D.W., and G.G. with input and approval from all authors. The content is that of the authors and is not intended to represent the official views of the National Institutes of Health.

DECLARATION OF INTERESTS

The authors declare no competing interests.

SUPPLEMENTAL INFORMATION

Supplemental information can be found online at <https://doi.org/10.1016/j.ymthe.2025.01.041>.

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